

Explicit Run–Return Enumeration for Rational Dyck Paths

From Pushdown Residuals to Weighted Binomial Determinants

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Abstract

Let $\mathcal{D}_{m,n}$ denote the set of $m \times n$ rational Dyck paths, with $m \geq n$. Dai, Fu, and Qiu recently introduced two rational run statistics, run and $\widetilde{\text{run}}$, and proved that each has a symmetric joint distribution with the return statistic ret . They left open the explicit enumeration of paths with prescribed run and return values. We give a finite determinant solution to both problems.

The key observation is that, in coarea coordinates $\mathcal{U}(P) = (u_1, \dots, u_n)$ with upper barriers

$$b_i = \left\lfloor \frac{m(i-1)}{n} \right\rfloor,$$

a return occurs precisely at an index for which $u_i = b_i$. Thus returns are upper-barrier contacts. If a finite set F of positive coarea values is forbidden, deleting the values of F from the integer line gives an order-preserving compression to a new barrier. A weighted Lindström–Gessel–Viennot determinant then enumerates paths avoiding F by their number of returns.

More generally, for $d \geq 1$ define the sampled-run statistic

$$\text{run}_d(P) = \min\{a \in [n] : da \notin \{u_1, \dots, u_n\}\}.$$

The ordinary run is $d = 1$, whereas the ratio-run of Dai–Fu–Qiu is $d = \lfloor m/n \rfloor$. Inclusion–exclusion over the required values $d, 2d, \dots, (a-1)d$ yields an explicit signed sum of binomial determinants for

$$\#\{P \in \mathcal{D}_{m,n} : \text{run}_d(P) = a, \text{ret}(P) = b\}.$$

We also eliminate coefficient extraction entirely: the joint count is written as two finite subset sums of integer binomial determinants, giving a coefficient-free explicit formula. The contact interpretation was suggested by a residual semantics arising in deterministic pushdown normalization for weighted-balance languages; after that observation, all proofs in the present paper are purely combinatorial. The published ordinary-run examples are recovered exactly, and ratio-run examples are displayed.

Keywords. rational Dyck paths; deterministic pushdown automata; weighted residual; run statistic; ratio-run; returns; coarea sequence; Lindström–Gessel–Viennot lemma; binomial determinant; inclusion–exclusion.

1 Introduction

Rational Dyck paths are lattice paths from $(0,0)$ to (m,n) , with north and east steps, that remain weakly above the line $y = nx/m$. They interpolate between classical Dyck paths, Fuss–Catalan paths, and the broader rational Catalan setting; see, for example, Armstrong, Loehr, and Warrington [1].

Dai, Fu, and Qiu [2] introduced two rational analogues of the classical run statistic. If

$$\mathcal{U}(P) = (u_1, \dots, u_n)$$

is the coarea sequence, their ordinary rational run is the first positive integer absent from the set of coarea values, while their ratio-run samples the arithmetic progression

$$\kappa, 2\kappa, 3\kappa, \dots, \quad \kappa = \left\lfloor \frac{m}{n} \right\rfloor.$$

They constructed composition-preserving involutions showing that both (run, ret) and $(\widetilde{\text{run}}, \text{ret})$ have symmetric joint distributions. Their Section 4 develops generating-function identities involving run and ret in the classical and Fuss–Catalan cases. In their concluding remarks they ask for an explicit formula for the number of rational Dyck paths with prescribed numbers of runs, of either type, and returns [2].

The present paper gives such formulas. Its route to the solution begins with the author’s earlier completion-aware deterministic pushdown work on weighted binary balance. The original object is a concrete 2:1 DPDA for the relation $N_1 = 2N_0$. Generalizing the arithmetic carried by that machine leads to the weighted residual

$$\rho_{p,q} = qN_1 - pN_0.$$

For a rational Dyck rectangle, writing $m = qg$ and $n = gp$ turns the usual path rank into the same quantity up to scale:

$$R = my - nx = g(qy - px) = g\rho.$$

Thus the Dai–Fu–Qiu return condition is a low-residual condition. In coarea coordinates this becomes the decisive contact identity

$$R_i < n \iff u_i = b_i, \quad b_i = \left\lfloor \frac{m(i-1)}{n} \right\rfloor.$$

Section 2 records this discovery chain explicitly, starting from the original DPDA itself and then passing to the general $p:q$ residual. The paper is not intended to obscure that provenance: the residual viewpoint is inherited from the preceding pushdown work. At the same time, no automata theorem is used as a hypothesis in the enumeration results; after the residual-to-rank bridge has been established, the proofs are self-contained combinatorics. In particular, the enumeration argument from Section 5 onward depends only on the rational-Dyck definitions, the return–contact correspondence of Theorem 4.1, and standard determinant/inclusion–exclusion tools.

Counting weakly increasing integer sequences below a barrier is classical. In particular, Kreweras gave determinant formulas for sequences between two barriers, and Pemantle and Wilf discuss the resulting binomial determinant for a single upper barrier [4, 5]. Our use of the determinant has a different refinement. Run conditions require specified *values* to be present or absent in the coarea sequence; they do not merely alter the barrier. We therefore first delete forbidden values from the integer line. The resulting order-preserving compression converts avoidance of an arbitrary finite set F into an ordinary barrier condition while preserving returns as selected upper contacts. A weighted nonintersecting-path determinant then marks those contacts. We use the Lindström–Gessel–Viennot principle in the binomial-determinant form developed by Gessel and Viennot [3].

The run constraints are then elementary. For every $d \geq 1$ we define

$$\text{run}_d(P) = \min\{a \in [n] : da \notin \mathcal{U}(P)\}.$$

Thus $\text{run}_1 = \text{run}$, while $\text{run}_\kappa = \widetilde{\text{run}}$. The condition $\text{run}_d(P) = a$ says that $d, 2d, \dots, (a-1)d$ all occur while ad does not. Inclusion–exclusion therefore converts the weighted forbidden-value determinant directly into the required joint enumeration.

Main contributions. The paper proves the following chain.

1. The pushdown residual $\rho_{p,q} = qN_1 - pN_0$ becomes the rational-path rank up to the factor g under $m = gq$, $n = gp$; hence returns lie in the low-residual band $0 \leq \rho < p$ (Section 2).
2. Returns are exactly upper-barrier contacts of the coarea sequence (Theorem 4.1).
3. Avoidance of an arbitrary finite set of positive coarea values is reduced by an order-preserving compression to a standard weakly increasing sequence below a new barrier (Lemma 5.1).
4. A weighted binomial determinant counts paths avoiding the forbidden set by their number of returns (Theorem 6.2).
5. Inclusion–exclusion yields a finite determinant formula for the joint distribution of $(\text{run}_d, \text{ret})$ for every $d \geq 1$ (Theorem 7.1). The cases $d = 1$ and $d = \lfloor m/n \rfloor$ give the two explicit enumerations requested in [2].
6. By row multilinearity of the determinant, coefficient extraction can be eliminated entirely, giving a coefficient-free formula as a finite signed sum of integer determinants (Theorem 7.4).

The symmetry theorems of Dai–Fu–Qiu are not used in the determinant derivation and therefore become independent identities satisfied by the explicit formulas obtained here.

2 From the original 2:1 DPDA to the $p:q$ residual

2.1 The original completion-aware 2:1 machine

The starting point is the completion-aware deterministic pushdown automaton shown in Figure 1. It was developed for the balance language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

Here the notation “2:1” refers to the symbol-count relation $N_1 : N_0 = 2 : 1$. The center/left/right core stores unresolved completion demand jointly in finite control and the stack. This machine is included because it is the concrete source from which the residual viewpoint used in the present paper was extracted.

A transition label $a, A/\gamma$ means that input a is consumed and top stack symbol A is replaced by γ ; the leftmost work symbol is the stack top. The symbol Z_0 is the bottom marker. The dashed boxes contain only right-endmarker completion operations and are not ordinary ε -moves during binary-input processing.

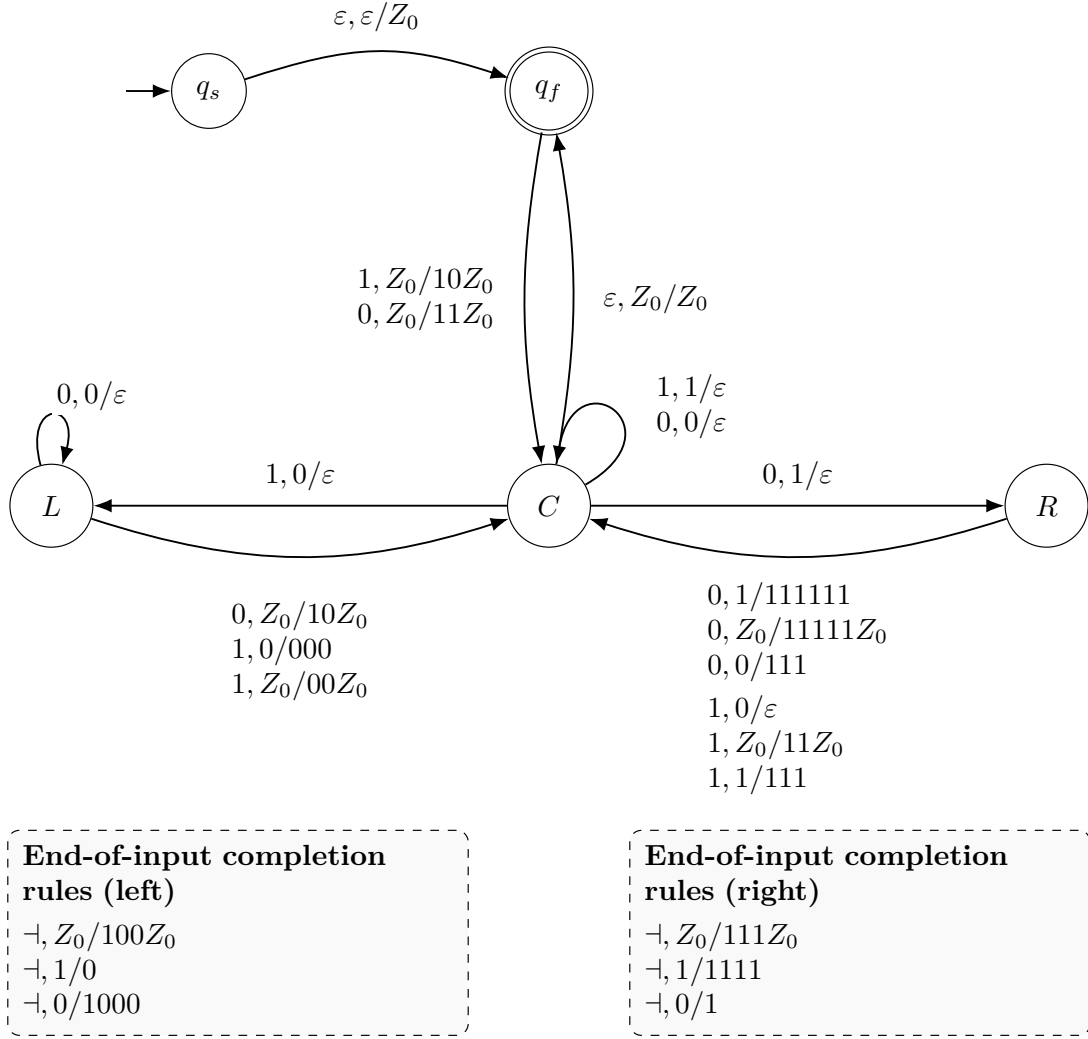


Figure 1: The completion-aware 2:1 DPDA from which the residual viewpoint was extracted. Solid arcs process binary input, while the dashed boxes contain only right-endmarker completion rules.

Ignoring the bottom marker, let $S \in \{0, 1\}^*$ denote the logical stack. For the 2:1 core define

$$D(S) = 2N_0(S) - N_1(S), \quad \Phi(C) = 0, \quad \Phi(L) = 3, \quad \Phi(R) = -3.$$

The processed input prefix has signed residual

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w), \quad (1)$$

and a direct check of the displayed binary transitions gives the distributed invariant

$$\rho_{2,1}(w) = D(S) + \Phi(q_{\text{state}}). \quad (2)$$

Thus the unresolved arithmetic is not tied to one particular stack word: part of the residual may be carried by finite control. For example, after the prefix 111 the core reaches (L, ε) , so the residual $\rho_{2,1} = 3$ is carried entirely by $\Phi(L) = 3$. This was the feature that survived the passage from the particular 2:1 machine to the general weighted-balance semantics.

2.2 From 2:1 to arbitrary coprime $p:q$

The machine-specific constants disappear when the balance relation is written in its general form

$$L_{p,q} = \{w \in \{0, 1\}^* : qN_1(w) = pN_0(w)\}, \quad \gcd(p, q) = 1. \quad (3)$$

The canonical scalar imbalance is

$$\boxed{\rho_{p,q}(w) = qN_1(w) - pN_0(w)}. \quad (4)$$

Reading a 1 changes ρ by $+q$, while reading a 0 changes it by $-p$. The primitive nonnegative zero-residual count packet is

$$K = (q, p),$$

meaning q zeros together with p ones. The original 2:1 DPDA is exactly the specialization $(p, q) = (2, 1)$ of this arithmetic viewpoint. The passage from that concrete machine to the scalar residual (4) is recorded here to make the provenance of the present method explicit. No automata theorem is used below; for the enumeration argument, only the scalar residual itself is needed.

2.3 From the $p:q$ residual to rational-path rank

Now pass from binary balance to a rational Dyck path. Encode an east step by 0 and a north step by 1. If a prefix ends at (x, y) , then $x = N_0$ and $y = N_1$. For the full rectangle let

$$g = \gcd(m, n), \quad m = gq, \quad n = gp, \quad \gcd(p, q) = 1.$$

The same scalar residual becomes

$$\rho = qy - px, \quad (5)$$

while the standard rational-path rank is

$$\boxed{R(x, y) = my - nx = g(qy - px) = g\rho}. \quad (6)$$

Because a rational Dyck path has $R \geq 0$, the Dai–Fu–Qiu return condition $R < n = gp$ is therefore equivalent to

$$\boxed{0 \leq \rho < p}. \quad (7)$$

This is the exact bridge used in the rest of the paper: the low-residual event suggested by the pushdown viewpoint becomes, in coarea coordinates, an upper-barrier contact. The latter statement is proved directly and independently in Theorem 4.1.

$$\boxed{\text{original 2:1 DPDA}} \longrightarrow \boxed{\rho_{p,q} = qN_1 - pN_0} \longrightarrow \boxed{R = g\rho} \longrightarrow \boxed{\text{return} = \text{coarea contact}}.$$

3 Rational Dyck paths and sampled runs

Throughout, m, n are positive integers with $m \geq n$. A path consists of north steps $N = (0, 1)$ and east steps $E = (1, 0)$.

Definition 3.1. An $m \times n$ rational Dyck path is a path from $(0, 0)$ to (m, n) that remains weakly above $y = nx/m$. Its set is denoted by $\mathcal{D}_{m,n}$.

For $P \in \mathcal{D}_{m,n}$, let

$$\mathcal{U}(P) = (u_1, \dots, u_n)$$

be its coarea sequence, where u_i is the horizontal coordinate of the i th north step, equivalently the number of cells to its left. Define

$$b_i := \left\lfloor \frac{m(i-1)}{n} \right\rfloor, \quad 1 \leq i \leq n. \quad (8)$$

Proposition 3.2 (Coarea characterization). *A sequence $u = (u_1, \dots, u_n)$ is the coarea sequence of a path in $\mathcal{D}_{m,n}$ if and only if*

$$0 = u_1 \leq u_2 \leq \dots \leq u_n, \quad u_i \leq b_i \quad (1 \leq i \leq n). \quad (9)$$

Proof. The weak inequalities express the fact that successive north steps occur in nondecreasing columns. Immediately before the i th north step the path is at $(u_i, i-1)$, which lies weakly above the diagonal precisely when

$$i-1 \geq \frac{n}{m}u_i,$$

or equivalently $u_i \leq m(i-1)/n$. Since u_i is integral, this is $u_i \leq b_i$. Finally $b_1 = 0$, so $u_1 = 0$. $\square$

Dai–Fu–Qiu define a return position as a nonterminal lattice point on the path having rank smaller than n , where the rank of (x, y) is $my - nx$ [2]. Their return statistic counts the corresponding north steps.

Definition 3.3 (Sampled run). For $d \geq 1$ and $P \in \mathcal{D}_{m,n}$, define

$$\text{run}_d(P) := \min\{a \in [n] : da \notin \{u_1, \dots, u_n\}\}. \quad (10)$$

The minimum exists because the sequence contains only n entries, one of which is 0.

For $d = 1$, Definition 3.3 is the ordinary rational run of [2]. If

$$\kappa = \left\lfloor \frac{m}{n} \right\rfloor,$$

then $d = \kappa$ gives their ratio-run $\widetilde{\text{run}}$.

4 Returns are coarea contacts

The following observation is the conceptual bridge used throughout the paper.

Theorem 4.1 (Return–contact correspondence). *For every $P \in \mathcal{D}_{m,n}$ with coarea sequence $(u_1, \dots, u_n)$,*

$$\boxed{\text{ret}(P) = \#\{i \in [n] : u_i = b_i\}}. \quad (11)$$

Proof. Immediately before the i th north step the path is at $(u_i, i-1)$ and hence has rank

$$R_i = m(i-1) - nu_i.$$

Write

$$m(i-1) = nb_i + r_i, \quad 0 \leq r_i < n.$$

Then

$$R_i = r_i + n(b_i - u_i). \quad (12)$$

By Proposition 3.2, $b_i - u_i \geq 0$. Therefore

$$R_i < n \iff b_i - u_i = 0 \iff u_i = b_i.$$

Every nonterminal path point of rank $< n$ must be followed by a north step: an east step would decrease the rank by n and make it negative, contradicting the rational-Dyck condition. Hence the north steps whose initial points have rank $< n$ are precisely the returns of [2]; the terminal endpoint is not an initial point of a north step. Summing over i gives (11). $\square$

Remark 4.2 (Pushdown residual interpretation). Let $g = \gcd(m, n)$ and write $m = gq$, $n = gp$ with $\gcd(p, q) = 1$. Under the step encoding $0 = E$, $1 = N$, a prefix containing y north steps and x east steps has the weighted-balance residual

$$\rho = qy - px,$$

which is the scalar residual obtained by generalizing the arithmetic carried by the original 2:1 pushdown machine. The rational-path rank is $R = g\rho$, so a return position satisfies

$$0 \leq \rho < p.$$

Thus Theorem 4.1 may equivalently be read as the statement that the finite low-residual band singled out by the pushdown semantics becomes upper-barrier contact in coarea coordinates. This remark records the origin of the viewpoint; the theorem itself depends only on the rank calculation above.

5 Deleting forbidden coarea values

Let $F \subset \mathbb{Z}_{>0}$ be finite. We wish to enumerate paths whose coarea sequence avoids F , while retaining the return weight from Theorem 4.1.

Define

$$\phi_F(t) := t - \#(F \cap [1, t]), \quad t \in \mathbb{Z}_{\geq 0}, \quad (13)$$

and

$$c_i(F) := \phi_F(b_i) = b_i - \#(F \cap [1, b_i]). \quad (14)$$

Also put

$$\epsilon_i(F) := \mathbf{1}_{\{b_i \notin F\}}. \quad (15)$$

Lemma 5.1 (Forbidden-value compression). *The map*

$$(u_1, \dots, u_n) \mapsto (v_1, \dots, v_n), \quad v_i = \phi_F(u_i), \quad (16)$$

is a bijection between

$$\left\{ \begin{array}{l} 0 = u_1 \leq \dots \leq u_n, \quad u_i \leq b_i, \\ \{u_1, \dots, u_n\} \cap F = \emptyset \end{array} \right\} \quad (17)$$

and

$$\{0 = v_1 \leq \dots \leq v_n : v_i \leq c_i(F)\}. \quad (18)$$

Moreover, whenever $b_i \notin F$,

$$u_i = b_i \iff v_i = c_i(F). \quad (19)$$

If $b_i \in F$, the equality $u_i = b_i$ is impossible on the domain.

Proof. For each $B \geq 0$, deletion of the elements of $F \cap [1, B]$ makes ϕ_F an order-preserving bijection

$$\{0, 1, \dots, B\} \setminus F \longrightarrow \{0, 1, \dots, \phi_F(B)\}.$$

Applying this coordinatewise with $B = b_i$ proves that (16) is well-defined, preserves weak order, and has a unique inverse obtained by reinserting the deleted values. This proves the bijection. If $b_i \notin F$, the maximum allowed value b_i maps to the maximum compressed value $c_i(F)$, proving (19). The final assertion is immediate. $\square$

6 A weighted binomial determinant

We use the convention

$$\binom{a}{b} = 0 \quad \text{if } b < 0 \text{ or } b > a,$$

for integers $a \geq 0$.

Definition 6.1. For a finite $F \subset \mathbb{Z}_{>0}$, define

$$D_F(z) := \det_{1 \leq i, j \leq n} \left[\binom{c_i(F) + 1}{j - i + 1} + (z - 1)\epsilon_i(F) \binom{c_i(F)}{j - i} \right]. \quad (20)$$

Theorem 6.2 (Weighted forbidden-value determinant). *For every finite $F \subset \mathbb{Z}_{>0}$,*

$$D_F(z) = \sum_{\substack{P \in \mathcal{D}_{m,n} \\ \mathcal{U}(P) \cap F = \emptyset}} z^{\text{ret}(P)}. \quad (21)$$

Proof. Write $c_i = c_i(F)$ and $\epsilon_i = \epsilon_i(F)$. Consider the directed acyclic lattice network with east steps $(1, 0)$ and northeast steps $(1, 1)$. For $1 \leq i, j \leq n$, set

$$S_i = (-c_i - 1, i), \quad T_j = (0, j + 1).$$

A path from S_i to T_j has horizontal displacement $c_i + 1$ and vertical displacement $j - i + 1$, hence there are

$$\binom{c_i + 1}{j - i + 1}$$

such paths.

If $\epsilon_i = 1$, give weight z to the northeast edge leaving S_i and weight 1 to all other edges. If $\epsilon_i = 0$, all edges leaving S_i have weight 1. The number of $S_i \rightarrow T_j$ paths using that first northeast edge is

$$\binom{c_i}{j - i},$$

so the weighted single-path generating function is exactly the (i, j) entry of (20).

By the Lindström–Gessel–Viennot lemma [3], the determinant is the signed generating function of vertex-disjoint path families connecting the sources to the targets. The source–target configuration is nonpermutable. Indeed, since c_i is nondecreasing, for $i < k$ we have $x(S_i) \geq x(S_k)$. On the common horizontal range from $x(S_i)$ to 0, an S_i -path starts strictly below an S_k -path. If their assigned targets were inverted, say T_j and T_ℓ with $j > \ell$, their vertical order at $x = 0$ would be reversed. Because every east or northeast step advances the horizontal coordinate by one, each path has one vertex at every integer abscissa in this common range; the change of vertical order therefore forces a common vertex. Thus a vertex-disjoint family has no inversion, and only the identity matching $S_i \rightarrow T_i$ contributes.

An identity path $\pi_i : S_i \rightarrow T_i$ contains exactly one northeast step and c_i east steps. Let $e_i \in \{0, \dots, c_i\}$ be the number of east steps occurring before its unique northeast step, and set

$$v_i := c_i - e_i. \quad (22)$$

For adjacent paths, vertex-disjointness is equivalent to

$$-c_{i+1} - 1 + e_{i+1} < -c_i + e_i,$$

which, since all quantities are integral, is equivalent to

$$c_i - e_i \leq c_{i+1} - e_{i+1},$$

that is, $v_i \leq v_{i+1}$. Thus identity nonintersecting families are in bijection with

$$0 \leq v_1 \leq \dots \leq v_n, \quad v_i \leq c_i.$$

This is precisely the compressed sequence class in Lemma 5.1.

Finally, π_i uses its first northeast edge if and only if $e_i = 0$, equivalently $v_i = c_i$. Such an edge is weighted by z exactly when $\epsilon_i = 1$. By Lemma 5.1, these weighted events correspond exactly to the original contacts $u_i = b_i$, and Theorem 4.1 identifies their number with $\text{ret}(P)$. Therefore each family has weight $z^{\text{ret}(P)}$, proving (21). $\square$

Corollary 6.3 (Return polynomial). *For $F = \emptyset$,*

$$\sum_{P \in \mathcal{D}_{m,n}} z^{\text{ret}(P)} = \det_{1 \leq i, j \leq n} \left[\binom{b_i}{j-i+1} + z \binom{b_i}{j-i} \right]. \quad (23)$$

Proof. When $F = \emptyset$, $c_i = b_i$ and $\epsilon_i = 1$. Pascal's identity gives

$$\binom{b_i+1}{j-i+1} + (z-1) \binom{b_i}{j-i} = \binom{b_i}{j-i+1} + z \binom{b_i}{j-i}.$$

Apply Theorem 6.2. $\square$

7 Explicit run–return enumeration

For $d \geq 1$ and $a \geq 1$, define the required sampled values

$$S_{d,a-1} := \{d, 2d, \dots, (a-1)d\}, \quad (24)$$

with $S_{d,0} = \emptyset$.

Theorem 7.1 (Main enumeration theorem). *Let $m \geq n \geq 1$, $d \geq 1$, and $a, b \in [n]$. Then*

$$N_{m,n}^{(d)}(a, b) := \#\{P \in \mathcal{D}_{m,n} : \text{run}_d(P) = a, \text{ret}(P) = b\} = [z^b] \sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} D_{T \cup \{ad\}}(z). \quad (25)$$

Proof. By Definition 3.3, $\text{run}_d(P) = a$ if and only if

$$S_{d,a-1} \subseteq \mathcal{U}(P) \quad \text{and} \quad ad \notin \mathcal{U}(P). \quad (26)$$

First restrict to paths avoiding ad . To force every value of $S_{d,a-1}$ to occur, apply inclusion–exclusion to the events that a specified value of $S_{d,a-1}$ is missing. For $T \subseteq S_{d,a-1}$, the corresponding class avoids the forbidden set $T \cup \{ad\}$ and therefore has return polynomial $D_{T \cup \{ad\}}(z)$ by Theorem 6.2. Thus the return polynomial of paths satisfying (26) is

$$\sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} D_{T \cup \{ad\}}(z).$$

Extracting $[z^b]$ gives (25). $\square$

Corollary 7.2 (Ordinary run–return). *For $a, b \in [n]$,*

$$\#\{P \in \mathcal{D}_{m,n} : \text{run}(P) = a, \text{ret}(P) = b\} = [z^b] \sum_{T \subseteq \{1, \dots, a-1\}} (-1)^{|T|} D_{T \cup \{a\}}(z). \quad (27)$$

Corollary 7.3 (Ratio-run–return). *Let $\kappa = \lfloor m/n \rfloor$. For $a, b \in [n]$,*

$$\boxed{\#\{P \in \mathcal{D}_{m,n} : \widetilde{\text{run}}(P) = a, \text{ret}(P) = b\} = [z^b] \sum_{T \subseteq \{\kappa, 2\kappa, \dots, (a-1)\kappa\}} (-1)^{|T|} D_{T \cup \{a\kappa\}}(z).} \quad (28)$$

Corollaries 7.2 and 7.3 give explicit finite formulas for the two joint enumeration problems posed in [2].

7.1 A coefficient-free form

Formula (25) is already finite, but the coefficient extraction can be removed. For a finite forbidden set F , define $n \times n$ matrices A^F and B^F by

$$A_{ij}^F := \binom{c_i(F) + 1}{j - i + 1} - \epsilon_i(F) \binom{c_i(F)}{j - i}, \quad (29)$$

$$B_{ij}^F := \epsilon_i(F) \binom{c_i(F)}{j - i}. \quad (30)$$

Then

$$D_F(z) = \det(A^F + zB^F). \quad (31)$$

For $I \subseteq [n]$, let M_I^F be the matrix whose i th row is the i th row of B^F if $i \in I$, and the i th row of A^F otherwise.

Theorem 7.4 (Coefficient-free explicit formula). *For $a, b \in [n]$,*

$$\boxed{N_{m,n}^{(d)}(a, b) = \sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} \sum_{\substack{I \subseteq [n] \\ |I|=b}} \det M_I^{T \cup \{ad\}}.} \quad (32)$$

In particular, the joint count is expressed using only finite subset sums, binomial coefficients, and integer determinants.

Proof. The determinant in (31) is multilinear in its rows. Choosing the zB^F contribution in exactly the rows indexed by I and the A^F contribution in all remaining rows gives

$$[z^b] D_F(z) = \sum_{\substack{I \subseteq [n] \\ |I|=b}} \det M_I^F.$$

Substitute this identity into Theorem 7.1. □

Remark 7.5. The formula is effective without symbolic polynomial expansion. For fixed m, n, d, a, b , every determinant in (32) is an $n \times n$ integer determinant. The two subset sums have respectively 2^{a-1} and $\binom{n}{b}$ terms before cancellations or structural simplifications.

8 Examples

We display joint matrices with rows indexed by the run value and columns by the return value.

Example 8.1 (Ordinary run, $(m, n) = (12, 4)$). With $d = 1$, Theorem 7.1 gives

$$\left(N_{12,4}^{(1)}(a, b)\right)_{1 \leq a, b \leq 4} = \begin{pmatrix} 52 & 30 & 8 & 1 \\ 30 & 8 & 1 & 0 \\ 8 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \quad (33)$$

This is the ordinary run–return matrix displayed in [2].

Example 8.2 (Ordinary run, $(m, n) = (13, 5)$). The formula gives

$$\begin{pmatrix} 106 & 106 & 39 & 9 & 1 \\ 106 & 39 & 9 & 1 & 0 \\ 39 & 9 & 1 & 0 & 0 \\ 9 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (34)$$

again reproducing the published example [2].

Example 8.3 (Ratio-run, $(m, n) = (11, 5)$). Here $\kappa = \lfloor 11/5 \rfloor = 2$. Corollary 7.3 gives

$$\boxed{\begin{pmatrix} 85 & 43 & 13 & 2 & 0 \\ 43 & 32 & 11 & 2 & 0 \\ 13 & 11 & 7 & 2 & 0 \\ 2 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}}. \quad (35)$$

For instance, there are exactly 11 paths with $\widetilde{\text{run}} = 2$ and $\text{ret} = 3$. The sum of all entries is 273.

Example 8.4 (Ratio-run, $(m, n) = (10, 4)$). Here $\kappa = 2$ and

$$\begin{pmatrix} 30 & 11 & 2 & 0 \\ 11 & 9 & 4 & 1 \\ 2 & 4 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \quad (36)$$

The total number of paths is 76.

The symmetry of the displayed matrices is guaranteed independently by the involutions of Dai–Fu–Qiu. In the present approach, that symmetry becomes a family of identities between the determinant sums in Theorem 7.1.

9 Computational verification

The proofs above are purely combinatorial and do not depend on computation. We nevertheless used exact computation as an audit of indexing, endpoint conventions, and the inclusion–exclusion signs. Two independent exact checks were performed:

1. direct enumeration of all weakly increasing coarea sequences satisfying (9), followed by direct computation of run_d and ret ;
2. independent evaluation of Theorem 7.1 from the forbidden-value determinants.

The displayed matrices in Section 8 agree entry by entry under these calculations. A small exhaustive determinant audit covered 33 sampled-run cases with no mismatches. An independent audit of the coefficient-free formula covered 86 cases with no mismatches. These computations are consistency checks only and are not used in place of the symbolic proofs.

10 Discussion

The main formula separates the solution into three transparent operations:

$$\boxed{\text{return} = \text{barrier contact}} \longrightarrow \boxed{\text{delete forbidden values}} \\ \longrightarrow \boxed{\text{weighted determinant} + \text{inclusion-exclusion}}.$$

The determinant technology itself is classical: the binomial determinant belongs to a well-established family associated with nondecreasing sequences below barriers [4, 5, 3]. The new step in the present solution is the reduction of the run constraints to forbidden-value avoidance while retaining return positions as weighted contacts of the compressed upper barrier; inclusion-exclusion then converts that refinement into the requested joint enumeration.

The sampled statistic run_d is used here as a unifying notation: it packages the ordinary case $d = 1$ and the ratio case $d = \lfloor m/n \rfloor$ into one theorem. No additional structural claim about intermediate sampling widths is needed for the explicit enumeration established here.

The automata-theoretic origin and the combinatorial proof play different roles in the paper. The completion-aware DPDA suggested that a weighted residual, rather than a particular stack representation, is the stable semantic quantity. Under the rational-path encoding this residual becomes the usual rank up to scale, and the low-residual band becomes the contact statistic of Theorem 4.1. No automata construction is needed after this bridge: forbidden-value compression, the weighted LGV determinant, and inclusion-exclusion provide the complete enumeration.

Statements and Declarations

Author contributions. Alp Eren Bütün: conceptualization, methodology, formal analysis, investigation, software, validation, visualization, writing—original draft, and writing—review and editing.

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Data availability. No external data sets were used in this study.

Code availability. No code is required to establish the mathematical results. Exact computational consistency checks are summarized in Section 9.

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